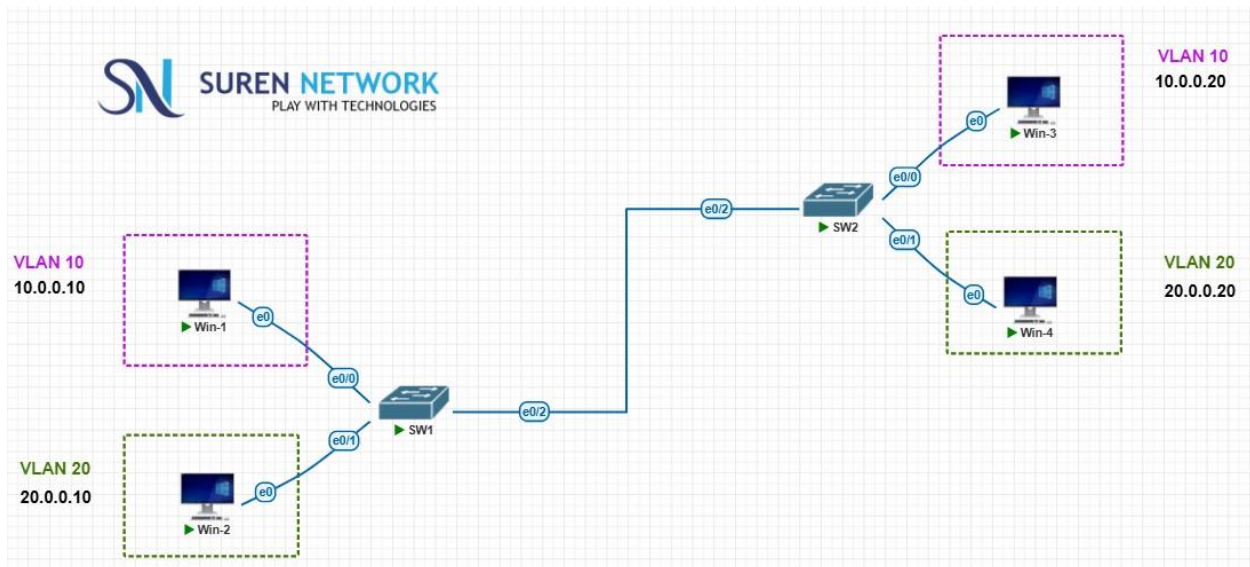


VLAN & TRUNK CONFIGURATION



Create vlan 10 and 20 in switch-1

```
SW1
SW1(config)#vlan 10
SW1(config-vlan)#name HR
*Sep 15 15:09:31.371: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Dis
g Wizard)
SW1(config-vlan)#name HR
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#vlan 20
SW1(config-vlan)#name SALES
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#
```

Move switchport to corresponding vlans

```
SW1
SW1(config)#int e0/0
SW1(config-if)#switchport access vlan 10
SW1(config-if)#exit
SW1(config)#
SW1(config)#int e0/1
SW1(config-if)#switchport access vlan 20
SW1(config-if)#exit
SW1(config)#
SW1(config)#
```

Check vlan database

```
SW1
SW1(config)#do sh vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Et0/2, Et0/3
10   HR                      active    Et0/0
20   SALES                   active    Et0/1
1002 fddi-default           act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
SW1(config)#
```

Enable trunk on ethernet1/2

```
SW1(config)#
SW1(config)#
SW1(config)#int e0/2
SW1(config-if)#switchport trunk encapsulation dot1q
SW1(config-if)#swutchoport mode trunk
^
% Invalid input detected at '^' marker.

SW1(config-if)#switchport mode trunk
SW1(config-if)#exit
```

Check trunk status

```
SW1(config)#do sh int trunk

Port      Mode      Encapsulation  Status        Native vlan
Et0/2     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Et0/2     1-4094

Port      Vlans allowed and active in management domain
Et0/2     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Et0/2     1,10,20
SW1(config)#
```

Create vlan 10 and 20 in switch-2

```
SW2
SW2(config)#
SW2(config)#vlan 10
SW2(config-vlan)#na
*Sep 15 15:11:14.110: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discover
g Wizard)
SW2(config-vlan)#name HR
SW2(config-vlan)#exit
SW2(config)#
SW2(config)#vlan 20
SW2(config-vlan)#name SALES
SW2(config-vlan)#exit
SW2(config)#
```

Move switchport to corresponding vlans

```
SW2(config)#
SW2(config)#int e0/0
SW2(config-if)#switchport access vlan 10
SW2(config-if)#exit
SW2(config)#
SW2(config)#int e0/1
SW2(config-if)#switchport access vlan 20
SW2(config-if)#exit
SW2(config)#
```

Check vlan database

```
SW2
SW2(config)#do sh vlan br
```

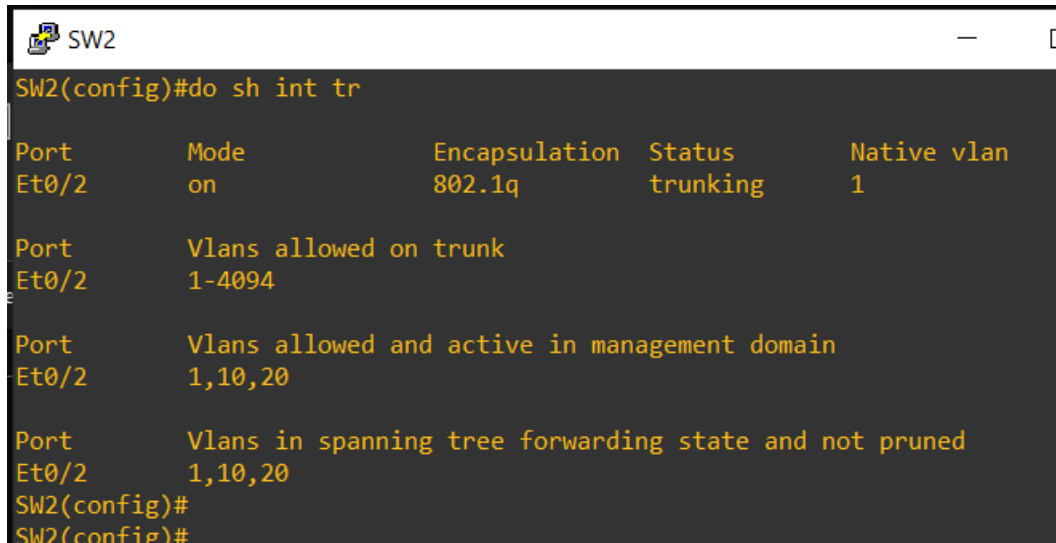
VLAN	Name	Status	Ports
1	default	active	Et0/3
10	HR	active	Et0/0
20	SALES	active	Et0/1
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
SW2(config)#
```

Enable trunk on ethernet1/2

```
SW2(config)#  
SW2(config)#int e0/2  
SW2(config-if)#switchport trunk encapsulation dot1q  
SW2(config-if)#switchport mode trunk  
SW2(config-if)#exit  
SW2(config)#
```

Check trunk status



```
SW2(config)#do sh int tr
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/2	on	802.1q	trunking	1

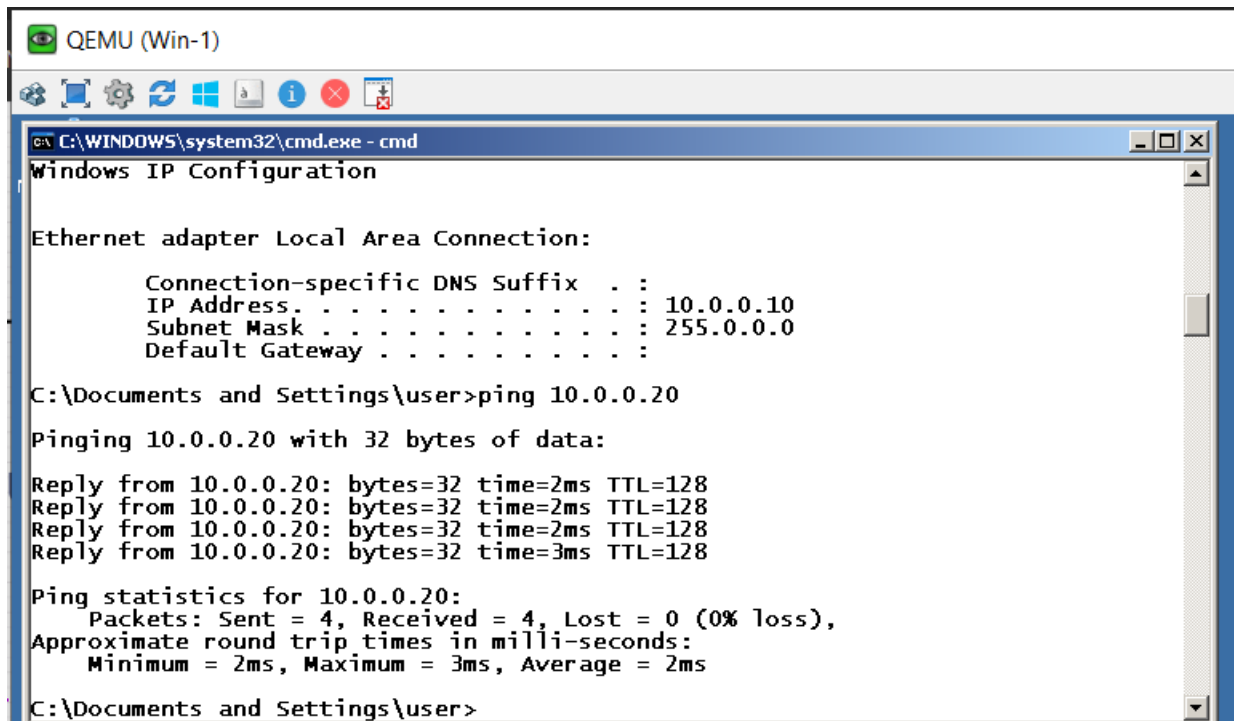
Port	Vlans allowed on trunk
Et0/2	1-4094

Port	Vlans allowed and active in management domain
Et0/2	1,10,20

Port	Vlans in spanning tree forwarding state and not pruned
Et0/2	1,10,20

```
SW2(config)#  
SW2(config)#
```

Connectivity Check



```
QEMU (Win-1)
```

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	
IP Address	:	10.0.0.10
Subnet Mask	:	255.0.0.0
Default Gateway	:	

```
C:\Documents and Settings\user>ping 10.0.0.20
```

Pinging 10.0.0.20 with 32 bytes of data:

```
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128  
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128  
Reply from 10.0.0.20: bytes=32 time=2ms TTL=128  
Reply from 10.0.0.20: bytes=32 time=3ms TTL=128
```

Ping statistics for 10.0.0.20:

Packets: Sent	= 4	Received	= 4	Lost	= 0 (0% loss),
Approximate round trip times in milli-seconds:					
Minimum	= 2ms	Maximum	= 3ms	Average	= 2ms

```
C:\Documents and Settings\user>
```

```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 20.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 

C:\Documents and Settings\user>ping 20.0.0.20

Pinging 20.0.0.20 with 32 bytes of data:

Reply from 20.0.0.20: bytes=32 time=3ms TTL=128
Reply from 20.0.0.20: bytes=32 time=3ms TTL=128
Reply from 20.0.0.20: bytes=32 time=1ms TTL=128
Reply from 20.0.0.20: bytes=32 time=2ms TTL=128

Ping statistics for 20.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```